BRAC University | CSE-460: VLSI Design | Spring 26 | Faculty:DPRN

Assignment - 01

Due - 11:59 PM, February 23 , 2026 | Submission link

ID:	Sec:	Name:

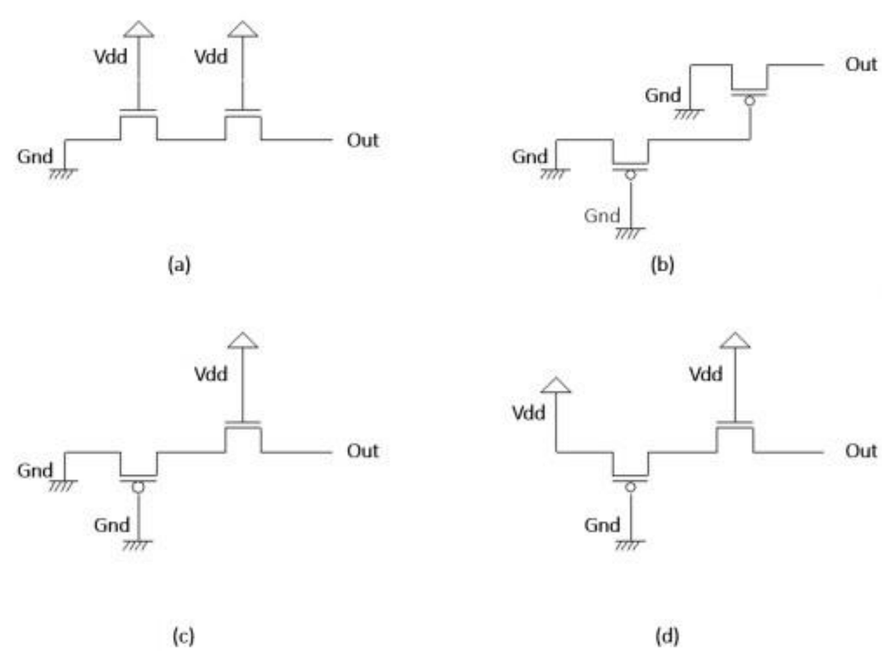
Instructions:

- **Print the question paper** and start writing your answers as instructed.
- **Do not print the questions on both sides of the pages**. You can **use the opposite side of the page** for writing your answers.
- You can add additional pages if required, but you **must write your ID, Section and Name at the top of each additional page.**
- Please note, **Collaboration =/= copying.** You are allowed to discuss the questions and clear confusions you might have, but you have to write your solutions independently and distinctively.
- **Scan** your answers, make a single PDF file by renaming “A3_StudentID_Fullname_CSE460.pdf” and **upload** it in the submission link.
- **Any findings of plagiarism will** result in a **Zero** mark. It may also hamper your future grading!
- **You must submit the online copy by the deadline.**

Q1. [CO3] - Marks 10

Design a CMOS compound gate that implements the following function:

Y = ((A . B . C) + D) . E



Determine the value of **Out** in all above conditions considering **Vdd = 3V**, **Vtp = -0.4V**, **Vtn = 0.5V** and **Gnd = 0V**.

ID:	Sec:	Name:

Q2. [CO3] - Marks 10

(CO1)	Consider you need to design a temperature control system for your classroom at BRAC University. Your system has 1 Output **D**, and 3 inputs **A**, **B** and **C** that come directly from three different sensors installed at the room. When a certain threshold of the temperature is crossed, each sensor gives a logical **HIGH** output. Your logic circuit should turn on the AC (**D = 1**) when the **majority** of inputs are **HIGH**. **[You can assume that you have the inputs available both in uncomplemented and complemented form]**		
	(a)	**Write** the truth table for the system.	
	(b)	**Determine** the boolean logic function of the system using KMAP and **design** the circuit. How many MOSFETs do you require to implement your design? Can you **design** the circuit using **NAND** gates only? **Show** how	

Q3. [CO3] - Marks 10

A D flip-flop is made by cascading two D latches as shown in Figure 3. Data input (**D**) and clock (**CLK**) for the circuit is shown in Figure 4. **Draw** the waveforms for Q_M and **Q**. (Use a pencil to draw the signals and draw on the question paper, no need to draw the diagram in your answer script separately.)

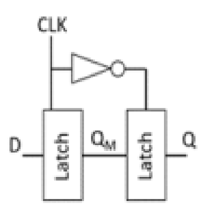


Figure 3

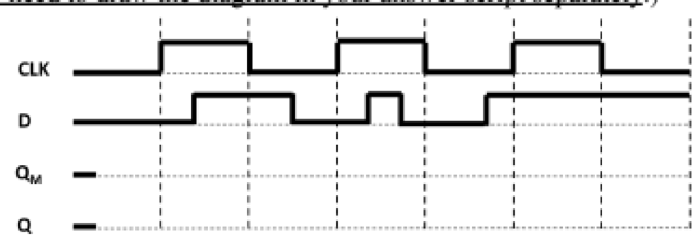


Figure 4

Identify the type of the flip-flop shown in figure 3.